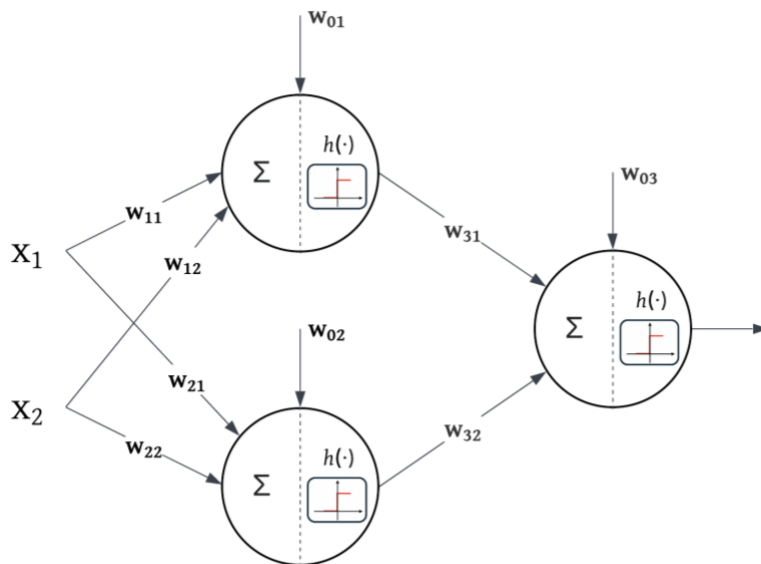


CS 767 Advanced Machine Learning Homework 7

Joseph Rissman

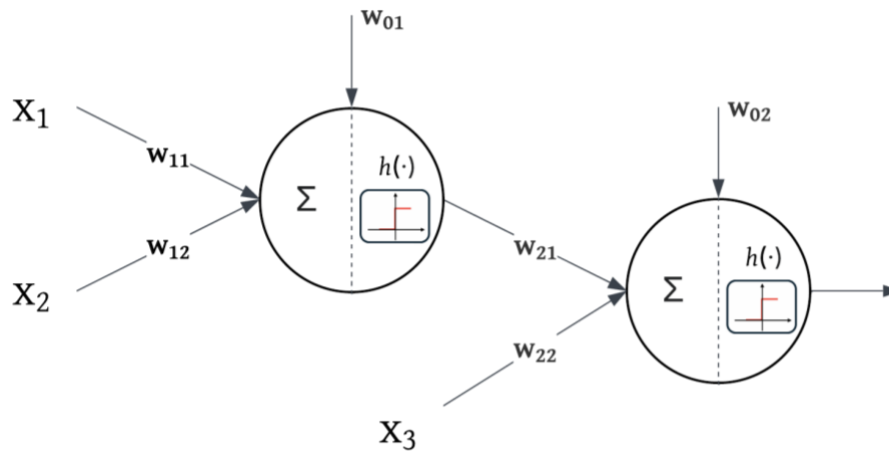
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1. **Logic Gates.** In class, we discussed that the XOR gate could be realized with a multi-layer feedforward network by observing that for any two binary inputs X_1 and X_2 , $X_1 \text{ XOR } X_2 = (X_1 \text{ AND } (\text{NOT } X_2)) \text{ OR } ((\text{NOT } X_1) \text{ AND } X_2)$. Draw the complete network for the XOR gate showing all weights and biases for each neuron in the network.

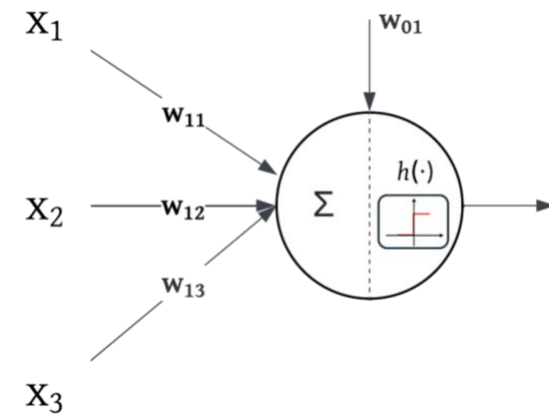


$$\begin{array}{lll}
 w_{11} = 1 & w_{12} = -1 & w_{01} = -0.5 \\
 w_{21} = -1 & w_{22} = 1 & w_{02} = -0.5 \\
 w_{31} = 1 & w_{32} = 1 & w_{03} = -0.5
 \end{array}$$

2. **Three-input gates.** Suppose I wish to implement X_1 AND X_2 AND X_3 . Can I realize this gate using (a) two neurons each with 2 inputs and a bias, and (b) a single neuron with 3 inputs and a bias? Whenever possible, draw the network with all weights and activation functions.



$$\begin{array}{lll} w_{11} = 1 & w_{12} = 1 & w_{01} = -1.5 \\ w_{21} = 1 & w_{22} = 1 & w_{02} = -1.5 \end{array}$$



$$w_{11} = 1 \quad w_{12} = 1 \quad w_{13} = 1 \quad w_{01} = -2.5$$

3. **Bagging.** In this exercise, we will numerically estimate the probability that a given observation is part of a bootstrap sample (**reference:** Sec. 5.2 and 5.4, ISLP).
- Generate a set $\mathcal{D}$ of $n = 1,000$ random Gaussian observations and tag observation $j = 4$. Generate 200 bootstrapped samples $\mathcal{D}^{(i)}, i = 1, 2, \dots, 200$, from this set and find the fraction of bootstrapped samples that do not contain the tagged observation.
 - Using the above procedure, create a plot that displays, for each integer value n from 100 to 100,000, the fraction of bootstrapped samples that contain the tagged (j^{th}) observation.
 - Given that $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$, comment on what your plot in Part b. above looks like in for large values of n .
4. **Boosting.** Complete the Scikit-Learn exercise on using the AdaBoost Classifier.